

DS 3001 : Exploratory Data Analysis

Q1

1) Show that $m(a+bX) = a + bX$

$$\begin{aligned} m(a+bX) &= \frac{1}{m(X)N} \sum_{i=1}^N (a+bx_i) \\ &\rightarrow \frac{Na + b \sum_{i=1}^N x_i}{N} \rightarrow a + bx_i = m(X) \end{aligned}$$

[$m(a+bX)$ ✓]

2) Show $\text{cov}(X, a+bY) = b \text{cov}(X, Y)$

→ $m(a+bY) = a + b m(Y)$

cov formula...

$$\text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) [(a+by_i) - (a+b m(Y))]$$

→ $a+by_i - a - b m(Y) \rightarrow b(y_i - m(Y))$

$$\begin{aligned} \rightarrow \text{cov}(X, a+bY) &= \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) b(y_i - m(Y)) \\ &\quad \text{take } b \text{ out} \\ &= b \left(\frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (y_i - m(Y)) \right) \end{aligned}$$

$$\text{cov}(X, a+bY) = b \text{cov}(X, Y) \quad \checkmark$$

3. Show $\text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X)$

$$= \frac{1}{N} \sum_{i=1}^N \left[(a+bx_i) - (a+b\bar{m}(X)) \right]^2 \rightarrow \begin{matrix} a+bx_i - a - b\bar{m}(X) \\ \rightarrow b(x_i - \bar{m}(X)) \end{matrix}$$

$$= \frac{1}{N} \sum_{i=1}^N \left[b(x_i - \bar{m}(X)) \right]^2 \rightarrow \text{take out } b^2$$

$$= b^2 \left[\frac{1}{N} \sum_{i=1}^N (x_i - \bar{m}(X))^2 \right] \rightarrow b^2 \text{cov}(X, X)$$

$\uparrow$ sample variance $X = S^2 = \uparrow$

$$= \text{cov}(a+bX, a+bX)$$

$$m(a+bX) = a + b\bar{m}(X)$$

4. Is $m(g(X)) = g(M(X))$ always true when g is non dec?

NO. $m(g(X))$: transform each, then average.

$g(M(X))$: avg. the original values, then transform

Ex: if $X = [0, 2]$ and $g(x) = x^2$

$$m(g(X)) = \frac{0^2 + 2^2}{2} = \frac{4}{2} = 2$$

$$g(M(X)) = \left(\frac{0+2}{2} \right)^2 = \frac{2}{2} = 1^2 = 1$$

$$m(g(X)) \neq g(M(X)) \because 2 \neq 1$$